# Optimal Design of Dual-Hop VLC/RF Communication System With Energy Harvesting

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***Abstract*— In this letter, we consider a dual-hop heterogeneous visible light communication (VLC)/radio frequency (RF) communication system to extend the coverage of VLC systems. Besides detecting the information over VLC link, the relay is able to harvest energy from the first-hop VLC link, by extracting the direct current component of the received optical signal, and uses the harvested energy to retransmit the data to a mobile terminal over the second-hop RF link. We investigate the optimal design of the hybrid system in terms of data rate maximization.**

***Index Terms*— Hybrid VLC/RF systems, energy harvesting, optimal design.**

## I. Introduction

VISIBLE light communications (VLC) had attracted a lot of attention as an indoor application of wireless optical technology [1], [2]. In VLC system, visible light of wavelengths of 380-750 nm is used as the medium for data transmissions. In this case, the light from a light emitting diode (LED) is used for the purpose of simultaneous illumination and data communication at high data rate [3]. However, the light originating from a LED source is naturally confined to a small area and is susceptible to blockage. As a solution for this problem, hybrid VLC/ radio frequency (RF) systems emerged in order to provide enhanced communication coverage [4], [5]. In [4], a number of VLC and RF access points are used to improve the coverage and the overall rate performance of the hybrid VLC/RF system. In [5], VLC is integrated with an RF-based wireless networks to improve the achievable data rates of mobile users.

In this letter, we introduce a dual-hop VLC/RF system as a new approach to extend the coverage of VLC systems. In this proposed hybrid VLC/RF system, the optical source, specifically a LED source, uses a subcarrier intensity modulation (SIM) scheme to transmit the information over a VLC channel to a relay. The relay uses direct detection (DD) scheme to retrieve the information. In SIM scheme, a pre-modulated information carrying RF signal is used to modulate the irradiance of a continuous optical wave after being properly direct current (DC) biased. This DC bias is added to ensure the optical intensity must satisfy the non-negativity constraint. The relay harvests energy from the received optical signal as long as the LED is on to illuminate the room. Then the relay uses the harvested light energy to retransmit its received information to a mobile terminal. The proposed dual-hop VLC/RF system provides a low complexity and energy efficient solution for coverage extension of VLC systems. In this letter, we propose two new methods for optimal DC bias design for the transmitted SIM signal. One method depends on the instantaneous second-hop RF channel power gain, while the other method depends on the average second-hop RF channel condition. The corresponding average end-to-end data rate of the system is analyzed for both cases. The first method was shown to give better average data rate performance over the second method. However, the preference of the first method comes at the cost of requiring instantaneous RF channel knowledge at the source. The major contributions in this letter are: 1) Novel dual-hop VLC/RF data transmission system with energy harvesting relay is introduced, 2) The energy carried by the DC component of the received SIM signal is harvested for data retransmission at the relay instead of discarding it as in conventional SIM/DD systems, and 3) The DC bias is optimized at the optical transmitter to maximize the overall transmission rate.

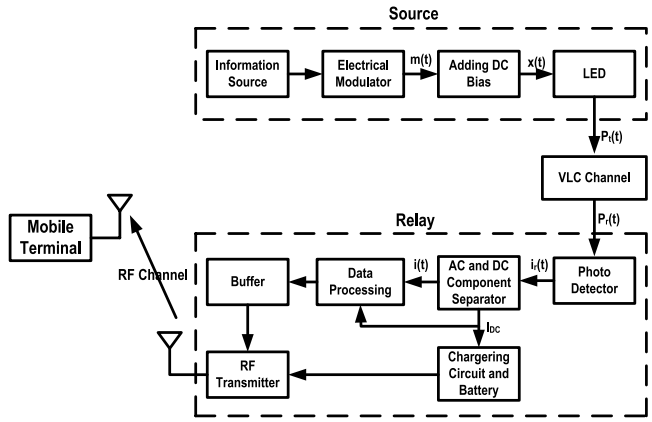


Fig. 1. Detailed block digram of the proposed system.

## II. Dual-Hop VLC/RF Transmission System

Assuming time division relaying scheme, in the first time slot of duration $T_1$, the information source terminal sends its signal over the VLC channel to the relay terminal. In the second time slot of duration $T_2$, the relay retransmits the source information to the mobile terminal over the RF channel. The relay harvests the light energy from the LED source during $T_1$ and $T_2$. Here, the relay is assumed to be located in a fixed position, where it has a clear line-of-sight communication link to the LED source. On the other hand, the designated user can move freely within the room. A detailed block digram of the proposed system is shown in Fig. 1.

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### A. Modeling of the LED Source to Relay Hop

The bit stream from the information source is first used to modulate an RF subcarrier to produce signal $m(t)$. A DC bias $B$ is added to the electrical modulated signal to insure that the resulting signal is non-negative, before being used to modulate the optical intensity of a LED array. At the same time, this DC bias places the LED in the proper operation mode. The transmitted optical signal can be written as $P_t(t) = NP_{LED}[B+m(t)]$, where $N$ is the number of LEDs in the LED array, and $P_{LED}$ is the single LED power assuming identical LEDs. At the relay, the received optical power is converted into electrical current through direct detection at the photo-detector. The electrical current $i_r(t)$ at the output of the photo-detector can be written as:

$$i_r(t) = \eta H P_t(t) + n(t) = I_{DC} + i(t) + n(t), \tag{1}$$

where $\eta$ is the photo-detector responsivity in $A/W$, $H$ is the VLC channel power gain, and $n(t)$ is an additive white Gaussian noise (AWGN) coming from background shot noise and thermal noise, with total variance $\sigma_1^2 = \sigma_{shot}^2 + \sigma_{thermal}^2$. The VLC channel power gain $H$ is given by $H = \frac{A_r(\xi+1)}{2\pi d_1^2}\cos^{\xi}(\phi)\cos(\psi)$ [7], where $A_r$ is the physical area of the photo-detector, $\xi$ is the Lambert index that depends on the semi-angle at half luminance $\phi_{1/2}$ as $\xi = -1/\log_2\cos(\phi_{1/2})$, $d_1$ is the transmission distance from the LED to the illuminated surface of the photo-detector, $\phi$ is the irradiance angle, and $\psi$ is the incidence angle.

*1) Information Detection at the Relay:* From (1), the AC component $i(t) = \eta H N P_{LED} m(t)$ carries the information. To avoid clipping distortion by the nonlinearity of the LED, the input electrical signal to the LED must be within the linear region of LED operation. This can be done by adjusting the DC bias current $B$ such that the resulting signal magnitude is always within the dynamic range of operation of the LED denoted by $D \triangleq I_H - I_L$, where $I_L$ denotes the minimum input bias current and $I_H$ denotes the maximum input bias current. In this case, the input electrical signal $m(t)$ varies around the DC bias $B$ with peak amplitude given by:

$$A = \begin{cases} B - I_L, & \text{if } B < \frac{I_L+I_H}{2} \\ I_H - B, & \text{if } B \geq \frac{I_L+I_H}{2}. \end{cases} \tag{2}$$

Thus, the received electrical power $P_r^e$ will be given by $P_r^e = (\eta H N P_{LED} A)^2$. It was shown in [8] that the shot noise is the dominant noise source in VLC channels. Therefore, the noise power is equal to $\sigma_1^2 \cong N_{shot} W = qIW$, where $N_{shot} = qI$ is the power spectral density of the shot noise, $I$ is the induced current in the photo-detector coming from ambient light, $q = 1.6 \times 10^{-19}$ is the electron charge, and $W$ is the effective noise bandwidth. The received electrical signal-to-noise ratio (SNR) at the relay, denoted by $\gamma_1$, can then be defined as:

$$\gamma_1 = \frac{(\eta H N P_{LED} A)^2}{\sigma_1^2}. \tag{3}$$

*2) Energy Harvesting at the Relay:* The maximum amount of energy that the relay can harvest during $T_1$ and $T_2$ is given by [9]:

$$E_h = 0.75 I_{DC} V_{oc}(T_1 + T_2), \tag{4}$$

assuming the same DC bias is used over $T_1$ and $T_2$. Here, $I_{DC} = \eta H N P_{LED} B$ is the DC component of the output current $i_r(t)$ from the photo-detector and $V_{oc}$ is the open circuit voltage of the photo-detector. $V_{oc}$ is related to $I_{DC}$ as $V_{oc} = V_t \ln\left(\frac{I_{DC}}{I_0} + 1\right)$ [9], where $V_t$ is the thermal voltage ($V_t \approx 25$ mVolt) and $I_0$ is the dark saturation current of the photo-detector ($I_0$ is in the order of $10^{-9} \sim 10^{-12}$ Ampere [10]). $E_h$ is upper bounded by $E_h = 0.75(T_1 + T_2)V_t I_{DC}^2/I_0$.

### B. Modeling of the Relay to Mobile Terminal Hop

In steady state operation, the detection and processing of the signal received over the VLC channel during $T_1$ consumes a portion of the total harvested light energy during previous $T_1$ and $T_2$ cycle. The remainder of this harvested energy is used for RF retransmission of the data during $T_2$. We assume that the energy used for data reception at the relay is negligible and, as such, the harvested energy is mainly used for data transmission. Thus, in steady state operation, the RF retransmission power is equal to $E_h/T_2$. Accordingly, the instantaneous received SNR at the mobile terminal, denoted by $\gamma_2$ can be defined as:

$$\gamma_2 = \frac{h_2^2 E_h/T_2}{G_2\sigma_2}, \tag{5}$$

where $h_2^2$ is the RF channel power gain with $E[h_2^2]$ normalized to unity, where $E[.]$ is the expectation operator, and $\sigma_2$ is the noise power assuming AWGN noise. $G_2$ is the second-hop path loss defined as $G_2 = (4\pi d_0/\lambda)^2 (d_2/d_0)^{\gamma}$, with $\lambda$ is the used RF carrier wavelength, $d_0 = 1m$ is the reference distance, $d_2$ is the distance from the relay to the destination terminal, and $\gamma$ is the path loss exponent (typically, $\gamma$ takes values 1.6 - 1.8 for indoor situations [11]). The RF fading gain $h_2$ is modeled by Nakagami-$m$ distribution. The probability density function (PDF) of $h_2^2$ is given by [12] as:

$$f_{h_2^2}(h_2^2) = \frac{m^m (h_2^2)^{m-1}}{\Gamma(m)} \exp\left(-m h_2^2\right), h_2^2 \geq 0, \tag{6}$$

where $m$ is the Nakagami-$m$ parameter indicating the fading severity over the second-hop RF channel. Accordingly, the received SNR $\gamma_2$ will have the following PDF and cumulative density function (CDF), respectively as [12]:

$$f_{\gamma_2}(\gamma_2) = \frac{\left(\frac{m}{\bar{\gamma}_2}\right)^m \gamma_2^{m-1}}{\Gamma(m)} \exp\left(-\frac{m\gamma_2}{\bar{\gamma}_2}\right), \quad \gamma_2 \geq 0, \tag{7}$$

$$F_{\gamma_2}(\gamma_2) = \frac{1}{\Gamma(m)}\gamma\left(m, \frac{m\gamma_2}{\bar{\gamma}_2}\right), \quad \gamma_2 \geq 0, \tag{8}$$

where $\bar{\gamma}_2$ is the average SNR of the second-hop RF channel given by $\bar{\gamma}_2 = E_h/T_2 G_2 \sigma_2$ and $\gamma(\cdot,\cdot)$ is the lower incomplete Gamma function defined in [13, eq. (8.350.1)].

## III. Average End-to-End Data Rate Analysis With Optimal VLC Bias Design

Suppose that the LED source transmits data to the relay with rate $R_1$ bits/second/Hz and the relay retransmits the data to the mobile terminal with rate $R_2$ bits/second/Hz. The data rate of the whole system will be limited by the smaller

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data rate between the two hops, given by $R = \min(R_1, R_2)$, where $R_1$ is given by the lower bound on the capacity of the VLC channel [14] as $R_1 = \log_2\left(1 + \frac{e}{2\pi}\gamma_1\right)$, and $R_2$ is given by the well-known Shannon formula as $R_2 = \log_2(1 + \gamma_2)$. We assume that the signals transmitted over two hops have the same bandwidth. It is clear that increasing the DC bias $B$ will lead to an increase in the amount of harvested energy at the relay which leads to an increase in the data rate of the second-hop RF channel $R_2$. However, to transmit more information over the first-hop VLC channel, we need to reduce the DC bias $B$ to increase the received electrical power $P_r^e$ and thus increase $R_1$. Thus, the optimal DC bias needs to be designed. In what follows, we present two methods to design the optimal VLC DC bias.

### *A. Optimal VLC Bias Design Based on Average End-to-End Data Rate (ADR-Based Method)*

Since the first-hop VLC channel is static channel in its nature and the second-hop RF channel is dynamic channel, then the average end-to-end data rate, denoted by $\bar{R}$, can be obtained by using the law of total expectation as (9) that appears at the bottom of this page, where $c = 2^{R_1} - 1$. To evaluate the integral part of (9), after the substitution with (7), we first express $\log_2(1 + \gamma_2)$ as $\ln(1 + \gamma_2)/\ln(2)$. Then, by applying the integration by part rule, and using [13, eqs. (2.33.10) and (8.352.7)], along with the binomial expansion rule, $\bar{R}$ can be expressed in this case in a closed form as (10) that appears at the bottom of this page, where $\Gamma(\cdot,\cdot)$ is the upper incomplete Gamma function defined in [13, eq. (8.350.2)]. The optimal bias which maximizes the average end-to-end data rate $\bar{R}$, denoted by $B_A^*$, can be obtained by solving $\frac{\partial \bar{R}}{\partial B} = 0$ for $B$. However, it is difficult to obtain $B_A^*$ in closed form. As an alternative method, we can treat the problem as one-dimensional optimization problem with uncertainty range of $[I_L, I_H]$, and use numerical methods, such as Golden-section search method [15], to search for the optimal $B_A^*$.

### *B. Optimal VLC Bias Design Based on Instantaneous End-to-End Data Rate (IDR-Based Method)*

To maximize the instantaneous end-to-end data rate $R$, we need to solve the following optimization problem:

$$\begin{aligned} &\underset{B}{\text{maximize}}\ R(B) = \min(R_1, R_2) \\ &\text{subject to } B \in [I_L, I_H]. \end{aligned}$$

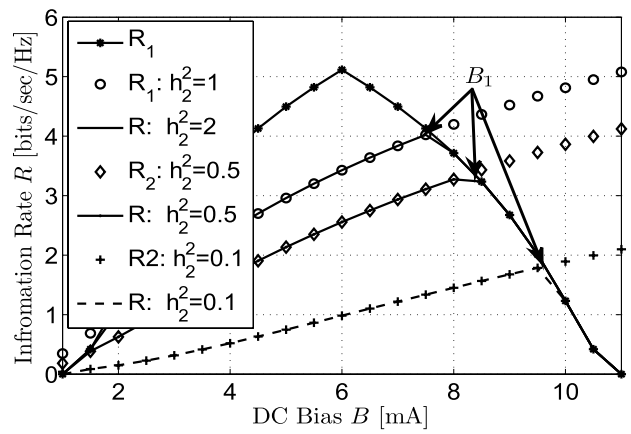


Fig. 2. System data rates with $d_2 = 10$m.

Normally, such max-min problem has solution at $R_1 = R_2$, which when we consider the constraint $B \in [I_L, I_H]$, leads to the solution:

$$B_1 = (c_1 I_H - \sqrt{c_1 c_2 I_H^2 h_2^2})/(c_1 - c_2 h_2^2), \tag{11}$$

with $c_1 = eG_2\sigma_2 T_2 I_0$ and $c_2 = 1.5\pi\sigma_1^2(T_1 + T_2)V_T$, where we plot $R_1$ and $R_2$ as function of $B$ in Fig. 2 using parameters given in section IV.

When second-hop RF channel gain $h_2^2$ increases (i.e., the quality of the second-hop RF channel improves), the intersection point of $R_2$ with $R_1$ (whose projection on the DC bias $B$-axis equals to $B_1$) moves up towards the peak of $R_1$ and thus the end-to-end data rate $R$ of the whole system increases with a decrease in the required DC bias $B$. However, the maximum end-to-end data rate is constrained by the peak value of $R_1$, which corresponds to the case when $B_1 = (I_L + I_H)/2$ (equals to 6 mA in Fig. 2) that occurs when:

$$h_2^2 \equiv h_{2_T}^2 = \frac{\left(\sqrt{c_1 c_2 I_H^2} - \sqrt{c_1 c_2 I_H^2 - 4c_1 c_2 c_m (I_H - c_m)}\right)^2}{4c_2^2 c_m^2}, \tag{12}$$

where $c_m = (I_L + I_H)/2$.

Thus, the optimal DC bias, denoted by $B_I^*$ in this case, can be chosen according to the following criteria:

$$B_I^* = \begin{cases} B_1, & \text{if } h_2^2 \le h_{2_T}^2 \\ \dfrac{I_L + I_H}{2}, & \text{if } h_2^2 > h_{2_T}^2, \end{cases} \tag{13}$$

$$\begin{aligned} \bar{R} &= E[\min(R_1, R_2)] = E[R_1 | R_1 \le R_2]\mathrm{P_r}[R_1 \le R_2] + E[R_2 | R_1 > R_2]\mathrm{P_r}[R_1 > R_2] \\ &= R_1[1 - F_{\gamma_2}(c)] + F_{\gamma_2}(c)\int_0^c \log_2(1 + \gamma_2) f_{\gamma_2}(\gamma_2) d\gamma_2. \end{aligned} \tag{9}$$

$$\bar{R} = R_1[1 - F_{\gamma_2}(c)] - \frac{F_{\gamma_2}(c)\left\{\ln(1+c)\Gamma\left(m, \frac{mc}{\bar{\gamma}_2}\right) - (m-1)!e^{\left(\frac{m}{\bar{\gamma}_2}\right)}\sum_{i=0}^{m-1}\sum_{k=0}^{i}\frac{(-m/\bar{\gamma}_2)^{i-k}}{k!(i-k)!}\left[\Gamma\left(k, \frac{m}{\bar{\gamma}_2}\right) - \Gamma\left(k, \frac{m(1+c)}{\bar{\gamma}_2}\right)\right]\right\}}{\Gamma(m)\ln(2)} \tag{10}$$

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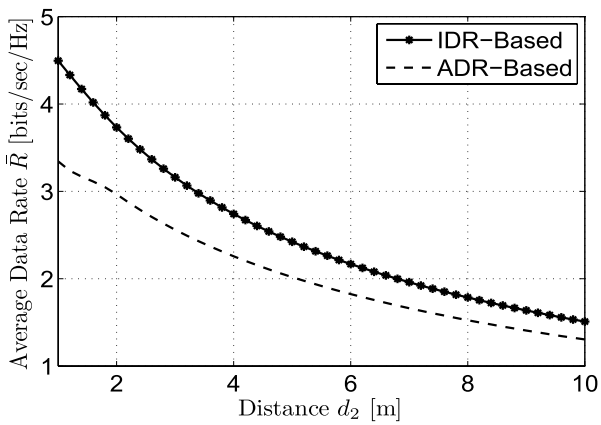


Fig. 3. System average data rate with optimal DC bias.

with optimum data rate $R(B_I^*) = R_1(B_I^*) = R_2(B_I^*)$. Accordingly, the average end-to-end data rate, with optimal DC bias $B_I^*$ chosen based on the instantaneous RF channel power gain $h_2^2$, is obtained by averaging the optimal data rata $R(B_I^*)$ over the distribution of $h_2^2$ as:

$$\bar{R} = \int_0^{h_{2T}^2} R(B_I^*) f_{h_2^2}(h_2^2) dh_2^2 + \int_{h_{2T}^2}^{\infty} R(B_I^*) f_{h_2^2}(h_2^2) dh_2^2. \quad (14)$$

The first integral part of (14) can be evaluated numerically as it is hard to be obtained in closed form. The second integral part of (14), can be given in a closed form as $\frac{\Gamma(m, m h_{2T}^2)}{\Gamma(m)} R_1|_{B_I^*=(I_L+I_H)/2}$ by using [13, eq. (3.351.2)].

## IV. Numerical Results

In this section, we present numerical examples to illustrate our analysis. We use $P_{LED} = 20\ mW/A$, $N = 10$, $\phi_{1/2}$=15 $deg.$, $\phi = 20\ deg.$, $\psi = 20\ deg.$, $d_1 = 1.5\ m$, $A_r = 0.04\ m^2$, $\eta = 0.4\ A/W$, $I = 5840\ \mu A$, $W = 1\ MHz$ [16], and $I_0 = 10^{-9} A$ for the VLC subsystem. For the RF subsystem, we use RF carrier frequency of 2.4 GHz, $\sigma_2 = 10^{-8}$, $\gamma = 1.6$, and $m = 1$. We use $T_1 = T_2 = 1 second$.

In Fig. 3, we plot the average data rate of the whole system as a function of the distance $d_2$ from the relay to the mobile terminal. For the ADR-based method, the RF channel path loss $G_2$ varies with $d_2$. For each value of $G_2$, we use the Golden-section search method with an accuracy of $10^{-6}$ to find the optimal DC bias $B_A^*$ and the corresponding average end-to-end data rate to obtain the ADR-based curve. For the IDR-based method, we use (14) to obtain the IDR-based curve. It can be seen from Fig. 3 that, the average data rate decreases as $d_2$ increases, using either ADR-based method or IDR-based method in choosing the optimal DC bias. However, the average end-to-end data rate had shown improvement when the optimal DC bias is chosen based on the instantaneous RF channel power gain $h_2^2$ (IDR-based method) over the method when the optimal DC bias is chosen based on RF channel path loss $G_2$ (ADR-based method), especially, at smaller distances between the mobile terminal and the relay. As the distance $d_2$ increases, the performance of the two methods approaches to each other. Note that, the performance advantage of the IDR-based method comes at the cost of requiring instantaneous channel knowledge of $h_2^2$ at the LED source to calculate the optimal DC bias $B_I^*$ according to the criteria given by (13).

## V. Conclusion

In this letter, we introduced a dual-hop heterogeneous data transmission system that composed of VLC and RF links. A relay between the two hops is able to harvest energy from the VLC link and uses it to retransmit the data to a mobile terminal over the RF link. The DC bias imposed on the electrical signal at the LED source affects both the power of the transmitted electrical signal over the first-hop VLC channel and the amount of energy harvested by the relay. Two methods of designing the optimal DC bias were introduced and the corresponding end-to-end average data rate of the system in each case was analyzed. Numerical results showed that the IDR-based method gives better average data rate performance over the ADR-based method. However, the performance advantage of the IDR-based method comes at the cost of requiring instantaneous second-hop RF channel knowledge at the LED source.

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